

Proposition Logic

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Proposition were given statements that results either 'True' or 'False'. Consider variables A, B, C,... which can have two values either 'True' or 'False'

K = 'Kritin is going to school' $\begin{matrix} \nearrow \text{True} \\ \searrow \text{False} \end{matrix}$

There are 3 logical operators:-

1. not ' , ' , ' $\neg A$
2. and ' $\wedge$ ' $A \wedge B$
3. or ' $\vee$ ' $A \vee B$

Negation or Complement $\begin{matrix} \text{not}(\text{not}(A)) = A \\ \text{not}(1) = 0, \text{not}(0) = 1 \\ \text{not}(A) = A' \\ \text{not}(A') = A \end{matrix}$

Associative $\begin{matrix} (A \text{ and } B) \text{ and } C = A \text{ and } (B \text{ and } C) \\ (A \wedge B) \wedge C = A \wedge (B \wedge C) \end{matrix}$

$(A \text{ or } B) \text{ or } C = A \text{ or } (B \text{ or } C)$
 $(A \vee B) \vee C = A \vee (B \vee C)$

would you like coke or orange juice
 $\begin{matrix} \uparrow & & \uparrow & & \uparrow \\ \text{operand} & & \text{operator} & & \text{operand} \\ \uparrow & & \uparrow & & \uparrow \\ A & & \vee & & B \end{matrix}$

I will not either take coke or orange juice. $= (A \vee B)'$

X = sky is clear

$\bar{X}$ = sky is not clear
 $\nwarrow$ operator

Implication or if... then $\begin{matrix} 'X' \text{ is true and } 'Y' \text{ is false, then the result is false,} \\ \text{otherwise it is true. It is represented by } \rightarrow / \Rightarrow \text{ symbol} \end{matrix}$

X	Y	$X \rightarrow Y$
0	0	1
0	1	1
1	0	0
1	1	1

POS

$$X \rightarrow Y = \bar{X} + Y$$

1	1	1
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1	1
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$X = \text{sky is clear}$
 $Y = \text{It is raining}$

$X \rightarrow Y = \text{if sky is not clear then it is raining}$
 $Y \rightarrow X = \text{if it is not raining then sky is clear}$

$A = \text{'You practice for exam'}$

$B = \text{'You will succeed'}$

$A \rightarrow B = \text{'If you not practice for exam then you will succeed'}$

$B \rightarrow A = \text{'If you will not succeed then you practice for exam'}$
 or
 $B \rightarrow A = \text{'If you will not succeed then you practice for exam'}$

if $\frac{5=7}{F}$ then $\frac{15=22}{F} \Rightarrow \text{True}$

if $\frac{x=2}{T}$ then $2x=4 \Rightarrow \text{True}$

if $\frac{x=2}{T}$ then $\frac{3x=5}{\text{Not possible}} \Rightarrow \text{False}$

if $x \neq 2$ then $\frac{3x=5}{\text{True}} \Rightarrow \text{True}$

if $\frac{5x > 15}{\text{false}}$ then $\frac{5x = 15}{\text{true}} \Rightarrow \text{True} \quad [\because x = 3]$